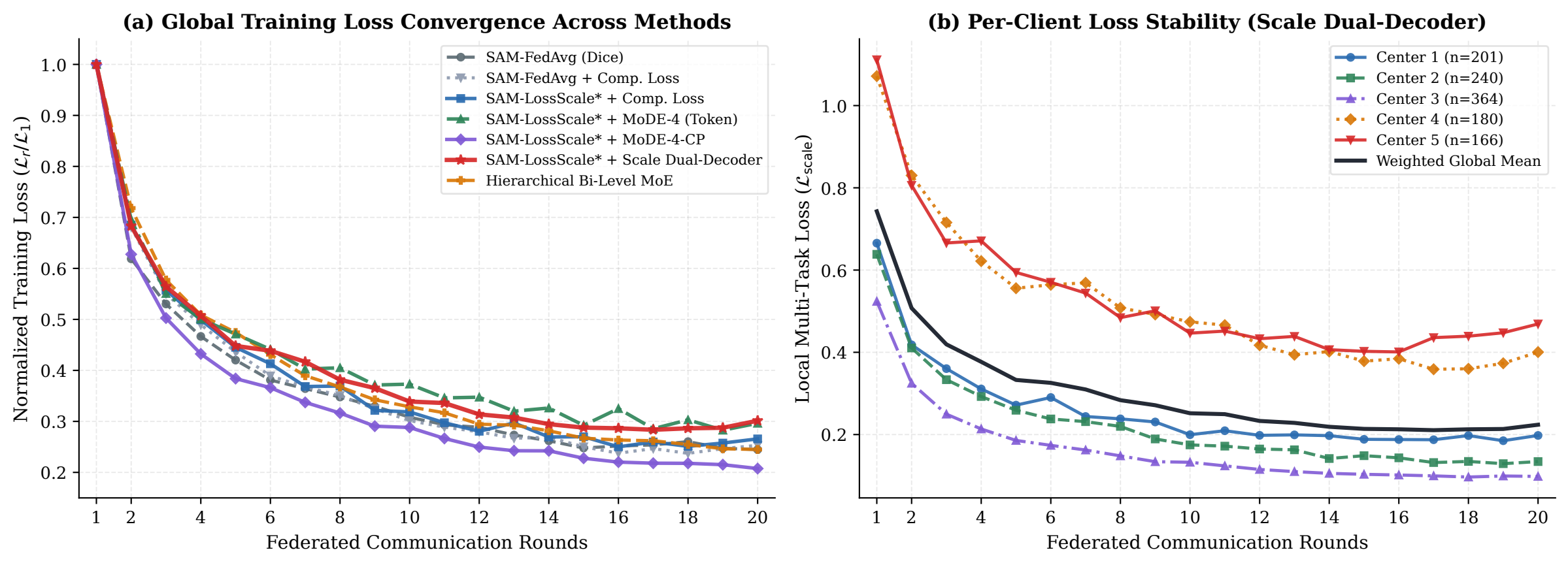
(a) Global Training Loss Convergence Across Methods
Normalized Training Loss ($\mathcal{L}_r/\mathcal{L}_1$)
Federated Communication Rounds
SAM-FedAvg (Dice)
SAM-FedAvg + Comp. Loss
SAM-LossScale* + Comp. Loss
SAM-LossScale* + MoDE-4 (Token)
SAM-LossScale* + MoDE-4-CP
SAM-LossScale* + Scale Dual-Decoder
Hierarchical Bi-Level MoE
(b) Per-Client Loss Stability (Scale Dual-Decoder)
Local Multi-Task Loss ($\mathcal{L}_{\text{scale}}$)
Federated Communication Rounds
Center 1 (n=201)
Center 2 (n=240)
Center 3 (n=364)
Center 4 (n=180)
Center 5 (n=166)
Weighted Global Mean